

Khawaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - II

Course Code : CSE 0533 1201 (CSE) & PHY 0533 1201 (EEE)

Term: Spring – 2024	Date: 24.01.2024	Time: 03.30 PM	Room: 327	Activity: Lecture 8
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Topics to be discussed: Entropy and Disorder

Target CLO & PLO : CLO1,2,3 & PLO1,5,9

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define Entropy & Disorder
- ❖ Explain the entropy in various thermodynamic process
- ❖ State the third law of the thermodynamics.
- ❖ Explain the term “Heat death of Universe”
- ❖ Solve assigned recommended mathematical problems

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion on some natural phenomenon	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session & Suggest for Assignment	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation & Assignment

Sample Questions (MCQ):

- 1) What is the unit of Entropy?
a) N b) J/K c) J d) Nm^{-1} e) J/Cal
- 2) Entropy
a) is decreasing b) is zero c) remain constant d) increasing e) is nothing

Sample Questions (SAQ):

- 1) What do you understand by Entropy? Is it having any unit?
- 2) “Adiabatic process is also called Iso-Entropy process”- explain it.
Or, Why the change of Entropy becomes zero in Adiabatic process?
- 3) Why does the entropy remain constant in reversible process?
- 4) Why does the entropy increase in irreversible process?
- 5) State the Zeroth law of the thermodynamics.
- 6) Explain the term “Heat death of Universe”
- 7) Find the change in entropy in order to change the temperature of 5kg water from 100C to 1000C.
[Specific Heat of water = 4200 J kg⁻¹K⁻¹]
- 8) If 4kg of water at 100°C is transformed into vapour of 100°C, find the increase in entropy. [Latent heat of vaporization of water = 2.26×10⁶ J kg⁻¹]
- 9) A heat engine takes 840J heat energy from the source and rejects heat energy of 630J to the cold sink. Find the efficiency of the heat engine.
- 10) Comments about Future of Universe from your concepts about Entropy in Thermodynamics.

Reference Books: 1. Giasuddin ahmed – Physics for Engineers (Part-1)
2. Halliday Resnick - Fundamental of Physics, 2nd edition

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 01-64
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 01-36